LaTeX Why Are Diffusion Model Images Not Watermarked Yet? Obstacles to large-scale Adoption and Resolution Approaches for Initial-Noise Watermarks in Latent Diffusion Models

Maria Jende[1]

[1]Film University Babelsberg KONRAD WOLF, Germany

Abstract

AI generated imagery poses growing risks of misinformation, deepfakes and copyright violations, making it complex for model owners, users and consumers to verify image origins. Invisible, initial noise altering watermarking for Latent Diffusion Models which require no model retraining, embedding the watermark before generation starts, has shown promise, yet large-scale application remains elusive. Therefore, to name the defining obstacles, six types of stakeholders and four major watermark properties (capacity, implementation and operation efficiency, imperceptibility and robustness) are formulated and analyzed. Followed by an evaluation of representative latent-noise methods (Tree-Ring [1], Gaussian Shading [2], T2SMark [3]) against this basis, revealing that current approaches redistribute trade-offs, meaning improvements in one parameter come at a cost in another, leaving robustness and efficiency as the decisive, unresolved properties. This examination shows that progressive refinement of existing paradigms have not yet closed the deployment gap and that reliable watermarking is requiring strong robustness of the model and watermark.

1. Introduction

Generative AI for synthetic image generation has become increasingly popular. While those models are supporting multiple commercial (e.g. in marketing or advertising) and private workflows (e.g. using Gen AI tools for entertainment and hobbies) [4], copyright as well as ethical concerns regarding the usage of synthetic images are becoming progressively urgent. For example, political mis- and disinformation are more than ever backed by the usage of GenAI imagery. Therefore, to secure democracy and trustworthy communication it is inevitable to be able to distinguish synthetic (GenAI images) from natural images (non-GenAI images). [5] Adding an invisible watermark to synthetic images, is one of the most recent, actively researched methods to help enable this distinction. [5–8] Due to their efficiency and output quality, generative image Diffusion Models have become, next to GANs, the most common approach for Gen AI Image models and therefore industry standard. [6] Large and popular models like *StableDiffusion*, *Midjourney* as well as *Dall-E3* and *Imagen* are diffusion models. [6]
Their generation method works by progressively diffusing image samples with Gaussian noise and then reverse that process to generate output images. [9] Invisible, state of the art watermarking methods for diffusion models utilize this set-up by altering the initial noise which is used for generation and embed a watermark in it. This approach offers several advantages to methods which are not modifying the sampling noise. [5–7]

Nevertheless, large-scale applicability of watermarks is an ongoing challenge which faces several obstacles. [5, 6, 8]
The urgency for watermarking described above raises the question of why AI watermarks are not yet in widespread use and what the current state of research in this field is. To understand what hinders widespread implementation of invisible watermarking in diffusion models, an analysis of the most influential parameters and functionalities of a watermark and their demands is conducted:

- **Capacity**: The amount of information that can be hidden with the watermark.
- **Implementation and Operation Efficiency**: The resources needed to integrate the method into a model.
- **Imperceptibility**: The impact on image generation diversity and quality.
- **Robustness**: The preservation of the watermark against removal and forgery.

Several methods address the impediments of these properties while balancing the trade-offs between them. Since initial noise-altering methods proved to be proficient [6, 8], three of them (Tree-Ring by Wen et al. [1], Gaussian Shading by Yang et al. [2] and T2SMark by Yang et al. [3]) that work in the latent space and progressively elevate each other, actively working on the described properties, are discussed below.

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.

2. Related Work

The rapid development of AI-generated imagery has led to significant research into watermarking techniques:

Zhao et al. [5] systematically review watermarking schemes for Artificial Intelligence Generated Content (AIGC), covering history, governance issues and current international regulations. Their formal taxonomy of attacks and desired properties for Gen AI watermarks comprises quality (distortion-freeness, undetectability), false-positive and -negative rates, robustness, unforgeability, message embedding and efficiency.
Their analysis of image watermarking methods, including Tree-Ring and Gaussian Shading, reveals inherent trade-offs that reinforce the obstacles selected in this paper.

Ren et al. [7] model AIGC watermarking as a node chain that maps out every point where a watermark can be embedded or attacked. They extend classical properties (e.g., robustness, capacity, stealthiness) with AIGC-specific ones such as endogeneity, plug-and-play capability and accessibility, and review attacks (tampering, removal, forgery) and governance in the US, EU and China, concluding with an urge for publicly detectable watermarks.

Luo et al. [6] focus solely on AI-generated images, surveying the evolution from traditional to deep-learning-based watermarking and discussing robustness against different attacks. They compare latent- and pixel-domain embedding and discuss fine-tuning-based methods, such as Stable Signature [10], and initial-noise methods, such as Tree-Ring, RingID [11] and Gaussian Shading, noting that latent-domain methods offer superior robustness at the cost of reduced capacity. I take up this observation by focusing on initial-noise-altering methods.

Zhao et al. [8] investigate regeneration attacks and expose a fundamental paradox: the same diffusion and VAE models that enable high-quality generation also serve as effective tools for watermark removal and reconstruction. They demonstrate that invisible pixel-level watermarks are removable, while semantic watermarks like Tree-Ring resisted their attacks, which is why I only consider initial-noise-altering methods. Since their work targets pixel-level perturbations and does not cover initial-noise watermarks, whether such watermarks can be deployed reliably at scale remains open.

While prior work surveys watermarking techniques [6], systematizes watermark properties and functionalities [5, 7] and tests vulnerabilities and attacks technically [6, 8], my analysis builds on it to review model users and their interests, identify four structural obstacles and evaluate the latest latent-noise methods against them, ultimately seeking to answer what hinders widespread adoption.

3. Diffusion Models

Text-to-image Diffusion Models, next to GANs and VAEs [6, 12], are the major method of synthetic image generation to this date due to their training stability and output quality and diversity. They operate by utilizing a bidirectional probabilistic framework consisting of a forward process (noise addition) and a reverse process (denoising) to generate images [12].
In the forward process (noise addition), the pixel distribution of a vast number of training images is gradually corrupted over a series of timesteps until it closely approximates a standard Gaussian distribution [1, 2].
To generate new samples, the model inverts the forward corruption by leaning on a Convolutional Neural Network (CNN), most commonly employing U-Net [13] or a Diffusion Transformer (DiT) [14] architecture [3, 6, 12]. It learns to reconstruct a clean sample by sequentially removing noise for a defined number of timesteps by predicting the underlying probability distribution of the training dataset and the noise added [1, 12].
The model evolves by backpropagation. During training, the loss between input and output is calculated resulting in a loss function which is progressively minimized to optimize the noise prediction network [6, 12]. During denoising, the CNN commonly applies either a Denoising Diffusion Probabilistic Model (DDPM) [9], or Denoising Diffusion Implicit Model (DDIM) [15], as sampling algorithm. Standard DDPM's rely on a stochastic Markovian-chain for the denoising process, leading to a slower multi-step sampling compared to DDIM's, which built upon DDPM's. They replace the Markovian-Chain with a class of non-Markovian, deterministic diffusion processes which allows for a more efficient sampling process due to its ability to skip timesteps and predict results [3, 6].
Most modern text-to-image systems like *Stable Diffusion*, *Imagen* or *DALL-E* are trained using DDPM but sample operating DDIM inversion or its successors (e.g., DPM-Solver, PNDM) to improve productivity [1, 2].
To further enhance efficiency, Latent Diffusion Models (LDM's) [16] reduce the computational complexity of Standard Diffusion Models while preserving sampling quality by training and sampling not in high-dimensional pixel space, but compressing the high-frequency details of the original image into a lower-dimensional latent space using an encoder.
The forward and backward processes, like DDIM, are then performed within this compact latent space, and a pre-trained decoder finally reconstructs the latent representation into pixel space [2].
The described models represent the state of the art in terms of architecture and efficiency, therefore, watermarking methods that focus on this described architecture were selected for analysis.

4. Types of Users

Different types of AI image generation model providers and users have different motivations and interests regarding the image usage and watermark functionalities. They can be condensed into six different roles:

(1) The model owner is the entity that trains and distributes the generative model, typically hosting it as non open-source system which can be accessed via APIs by users [2, 17]. One of them being *(2) the compliant user*, who is accessing the models services while adhering to platform policies and community guidelines [2, 7]. *(3) The malicious user*, on the other side, is as well rightfully accessing the model via an API but with the motivation to generate deepfakes, misinformation or illegal content [2, 3, 7].
Opposite to the first two user types, *(4) the unauthorized user* is not using the owners services directly but illegitimately obtains generated images to claim copyright or ownership over them [2, 3]. Both the malicious and unauthorized user are likely to become *(5) ad-*

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.

Stakeholder	Watermark Property			
	Capacity	Efficiency	Imperceptibility	Robustness
Model Owner	✓	✓	✓	✓
Compliant User	✓	✓	✓	✓
Adversary *(Malicious & Unauthorized User)*	–	–	–	✗
Consumer	✓	–	–	✓

✓ Active interest or need for property ✗ Performs active removal or forgery attacks on property – Passive or no need for property

Table 1: *Overview of watermark properties and their relevance for each stakeholder.*

versaries to the watermark by actively attempting to either remove watermarks to distribute harmful content anonymously, potentially evading accountability, or forge [1, 7] watermarks to falsely attribute synthetic or illegal content to a victim, causing defamation and reputation damage [8, 17].

A seldomly discussed, but omnipresent entity when it comes to safeguarding trustworthiness of imagery is *(6) the consumer*. They are not directly using the generation model to create but are consciously or unconsciously consuming their products and might actively or passively be influenced by their presence e.g. in forms of mis- and disinformation [8] or becoming a victim to deepfake generation. It can be assumed that those individuals share an interest in watermarks help distinguish between synthetic and AI generated content as well as who can be held accountable for creating and distributing them [8].

5. Watermark Properties and Their Implications for Large-Scale Implementation

The described needs and motivations of the individuals described above lead to a set of core properties and functionalities which an invisible synthetic image watermark should provide. Since the structure and importance of each property varies depending on the specific use case and watermarking approach, no single configuration is universally optimal [6, 8]. Still, several core parameters can be selected and the following section discusses the most common and practically relevant properties along with their advantages and disadvantages and potential impact on large-scale application. Table 1 presents this analysis in the context of individual stakeholder interests in a simplified overview.

5.1. Capacity

A watermark, depending on its capacity, aims to at least answer the binary question if an image is synthetic or natural. This is provided by a single-bit watermark [2,6]. Additionally, both the model owner and those of the compliant users who see themselves as content creators as well as consumers or victims of misuse, share an interest in the watermark being able to carry additional information, e.g. the model used for generation or the specific user that generated the image, allowing to trace back the entity owning and creating the image. This is offered by multi-bit watermarks, which can be, when assigned to individual API accounts, embed information of the used model and creator [6]. However, those user-specific IDs in generated images also hold data privacy risks: Embedding user data without user agreement can lead to critical privacy violations, e.g. if they are de-anonymised without consent. [5].

5.2. Implementation and Operation Efficiency

The model owner especially has an interest in keeping the model's resource consumption as low as possible [5]. Assumably, this is also an indirect interest of the users since high company expenses typically mean higher pricing of their services.

A watermark that requires a large amount of resources - like energy, time and manpower - to implement and operate on a long term are not attractive for model providers and therefore are not likely to be implemented in a large-scale, especially if there are no legal requirements for them to do so [5, 7].

Methods requiring model retraining or fine-tuning have a high integration threshold, especially as evolving large models demand adaptable watermarking methods [6, 7].

Beyond the computation cost, different model architectures pose a further challenge for large-scale application. For example, many noise-based watermarking schemes depend on ordinary differential equation (ODE)-based samplers [18] to reconstruct the initial noise during detection, meaning a change in the model's inference setup would make watermark recovery impossible [2, 3].

Therefore, methods that leverage a plug-and-play logic and efficiently adapt to different model architectures have a lower integration threshold and hold potential for long-term implementation.

5.3. Imperceptibility

This impact on model performance and resource consumption links closely to a watermark needing to be imperceptible. Imperceptibility ensures the watermark is invisible to the human eye and only machine readable, while it does not degrade the visual quality of the generated content [6, 7, 12]. Methods that alter the initial noise distribution of the sampling data, as is discussed in Section 6, might have a negative impact on first, the invisibility of the watermark [6, 12, 17] and secondly, the sampling process itself which might lead to a degradation in image quality and distortion of visual diversity [2, 3, 5]. Those potential compromises on model performance and generative quality are certainly nothing economically competitive companies as well as users would want to forgo [2].

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.

Especially if the effectiveness and reliability of a watermark is not guaranteed, because ensuring robustness while preserving generation quality of a watermark remains a challenge [3, 7].

5.4. Robustness

Foremost, robustness refers to the watermark's ability to survive different kinds of image transformation and geometric attacks [5, 6]. Those can be non-malicious, like side effects of post-processing operations, such as compression, cropping and color enhancement. On the other side, malicious attacks actively want to destroy the watermark, utilizing different methods, for example regeneration attacks, where the watermark is removed by first partially destroying the image including the watermark and then reconstructing it to regain image quality [1, 6, 12].
Unfortunately, not just the illegitimate removal of a watermark poses a problem, but also the forging of it. Hereby an attacker forges a new watermark without modifying the model, creating ownership ambiguity that may be unresolvable [5–7].

In removal and forging, the technical advancements of the generation models themselves help adversaries eliminating or faking them, making the generation model become the threat model itself. If the model is not robust enough against modification, an adversary can use a proxy model to invert the generation process and recover the watermarked noise vector: By sifting out the systematic distribution shifts because of the watermark implementation in generated images, which reveal information about the watermarking scheme [3, 17]. This information can then be used to generate new forged images that carry a valid watermark.
Open source models watermarking schemes are even more vulnerable as users can employ various custom decoding methods to bypass or strip the watermark [5].

The ultimate goal of the attacks and forging is tricking the watermark detector. Detection is not a parameter of the watermark itself, but for the application of its purpose and is therefore indispensable. Thereby the detector inherits the important purpose in verifying if an image is AI generated or not and eventually recovering additional information [5].
An effective detector has a fairly low false-positive/-negative rate when detecting the presence and content of a watermark. A high false-negative rate allows AI-generated content to bypass detection, making the detector ineffective, e.g. for copyright protection or content traceability. A high false-positive rate is untenable as it destroys the credibility of the detection system and leads to false accusations, feeding into the unreliability issue [5, 6].
The detector capabilities are therefore inherently intertwined with the robustness property, whose purpose is to ensure detectability by preservation.

6. Initial-Noise Watermarks in Latent Diffusion Models

There are several key challenges that hinder widespread implementation of watermarks in diffussion models.
First, the unreliable credibility of the watermark due to a lack of robustness against attacks and forging as well as detector failure, because a watermarking method that offers only empirical quality promise is less suitable for large-scale implementation [5]. By improving the robustness against attacks and forgery as well as potential model modification, a watermark method can improve their reliability and consequently their potential for implementation.
Secondly, the required resources and potential impact on generation quality make it unappealing or too inefficient for model owners to commit to watermarking their models.
Several watermarking methods adress those obstacles in different ways. The following analysis is looking at three approaches: Tree-Ring (TR) by Wen et al. [1], Gaussian Shading (GS) by Yang et al. [2] and T2SMark (T2SM) by Yang et al. [3].
They operate in the latent space of an LDM like *Stable Diffusion,*, which is the model all three papers evaluate on and are altering the initial noise to embed the watermark.
By mutually reinforcing each other they try to conquer the trade-off of their predecessor's method, sharing an identical technical set-up: Since they are offering a flexible plug-in logic, all three approaches do not require retraining or fine-tuning of the model. They focus on latent-based diffusion models which utilize only (GS, T2SM) or mainly (TR) ODE-based DDIM-inversion [15] for detection, causing all methods to be not (GS, T2SM) or only to a limited extent (TR) compatible with other sampling algorithms. Therefore, if DDIM exertion declines, the methods would need to be re-engineered [1–3].
The embedding happens at the initial latent noise vector before generation and is diffused across the entire semantics of the image during denoising. This poses a great advancement in robustness since the watermark is actively propagated and mixed across all spatial positions of the latent noise vector at every denoising step [1–3]. Different from post-hoc pixel-space watermarking, where the signal is an additive perturbation applied after the image is fully generated, initial noise watermarks are a structural property of the image, recoverable only by inverting the diffusion process back to the noise domain [6, 17].

The main difference between the methods lies in how they alter the latent space vector of the initial noise. Their individual approaches and trade-offs are discussed below.

6.1. Tree-Ring

The foundational Tree-Ring Watermark method by Wen et al. [1] embeds a single-bit watermark pattern into the Fourier transform of the initial noise vector, rather than into the noise vector directly.
The Fourier transform decomposes the image into its frequencies, describing how pixel values change across the image space in the frequency domain.
Through a circular mask in Fourier space the selected frequency components are replaced with a fixed key while the rest of the noise remains the initial Gaussian noise. The entire sampling process is influenced by this watermarked noise, increasing the watermarks robustness to image transformations like cropping and rotation as well as removal attacks. The TR can be implemented like a plug-in and requires a single pre-generation Fourier transform operation on the latent noise vector. This poses a major progress to prior methods, which were focusing on the generation output rather than on the generation process itself by embedding the watermark after generation or required retraining or fine-tuning of the model.
However, the robustness improvement goes along with a trade-off

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.

	Strength/Improvement	Trade-Off/Unresolved	FID	CLIP Score	TPR (C/A)	Det. Acc. ↓	Divers. ↑
TR	Robustness	Imperceptibility ǀ Capacity	55.9084 ± 0.53	0.3493 ± 0.0005	0.878/0.318	0.984	0.5924
GS	Imperceptibility ǀ Capacity	Robustness ǀ Efficiency	56.2020 ± 0.67	0.3502 ± 0.0004	**1.000/0.990**	0.991	0.5176
T2SM	Imperceptibility ǀ Robustness	Robustness	**55.8121 ±0.49**	**0.3499 ±0.0004**	1.000/0.985	**0.518**	**0.6102**

Table 2: *Each watermarking method shows individual trade-offs across watermarking properties, which the left part provides a comprehensive overview of. This follows performance evaluation results obtained using the Stable Diffusion 3.5 Medium model (SD3.5) (DiT architecture with a 16-channel latent space), obtained from T2SM. The Fréchet Inception Distance (FID) assesses the overall quality and fidelity of the generated images, while the Contrastive Language–Image Pre-training (CLIP) Score evaluates the alignment between the generated images and their corresponding text prompts. Both metrics are shown as means with their standard error (±) and, like Diversity, are judged by how closely they match the values of the unwatermarked SD3.5. The True Positive Rate (TPR) measures the detection performance by comparing clean and attacked images (C/A). Detection Accuracy (Det. Acc.) quantifies the imperceptibility of the watermark relative to unwatermarked images, while Diversity (Divers.) measures the visual variety of the generated images.*
An ↑ next to an evaluation metric indicates that higher values are preferable, whereas a ↓ indicates that lower values are preferable.

in generation quality: It adds structural change to the noise distribution by overwriting the frequency components inside the masked region. The larger the radius of the pattern, the higher is the robustness but also the disruption of initial noise distribution. GS experiments demonstrate that it can introduce indirect quality degradation, including prompt inconsistencies and limited sampling randomness [2].
Different from subsequent methods (GS and T2SM), TR does not restrict itself to LDMs but can also be integrated into pixel-space diffusion models. Therefore it can be applied to a wider range of architectures, but, because of the single-bit set-up, no additional generation information is embedded and capacity remains an open question.

6.2. Gaussian Shading

Gaussian Shading by Yang et al. [2] adopts and develops further Tree-Ring's approach of embedding the watermark directly into the initial noise by encrypting a multi-bit watermark and mapping it into the entire initial latent noise vector during image generation, aiming to improve the generation quality trade-off and capacity of the watermark.
The mapping is set-up so that the watermarked noise vector is statistically indistinguishable from the standard Gaussian distribution, posing a solution to TR's noise modification difficulty. In addition, different to TR, GS is leveraging the entire noise vector instead of a masked region to embed the key. They explicitly rule-out frequency domain manipulation because they argue that TR's Fourier approach disrupts the Gaussian distribution which leads to degraded generative quality. On the other hand, when working in the frequency domain, watermark robustness against geometric attacks such as rotation or cropping is strengthened, because the spatial relationship between the noise pattern and the image content remains more recognizable even after geometric transforms. The same applies to perturbation attacks, like addition of Gaussian Noise or compression, which are not directly editing the pixel position but their values: Frequency domain watermarks, like TR, can be designed to sit in lower frequency regions which survive smoothing and compression better.
Furthermore, GS enables user and model traceability due to the multi-bit watermark, which allows additional information embedding, and further develops TR's open question of user traceability.
Even though the authors of GS claim almost no performance loss, the analysis T2SM suggests that different keys do not perform equally and produce different fixed patterns inside the noise, causing inconsistent generation quality. They found that GS's Fréchet Inception Distance (FID) score, which evaluates the generation quality, fluctuates significantly depending on what particular watermark key combination is in use [3].
Furthermore, Arabi et al. notes that using the same initial noise across multiple generations is not truly distortion-free when examining the generated images as a group and is impacting generation diversity [17].

6.3. T2SMark

Like Gaussian Shading, T2SMark by Yang et al. [3] builds on the solid foundation of prior methods, trying to improve their trade-offs, focusing especially on robustness improvement while preserving the generation quality improvement developed by GS.
Its core innovation is Tail-Truncated Sampling (TTS), which leverages the natural value distribution of a standard Gaussian Noise. A standard Gaussian distribution assigns high probability to positive as well as negative values near zero (center-region) and low probability to values far from zero (tail-regions). So, rather than mapping watermark bits across the full Gaussian distribution of the noise vector, how it is done in Gaussian Shading, TTS divides the Gaussian distribution into three zones to address the vulnerability of near-zero Gaussian values: A bit-0 tail region (negative values), a bit-1 tail (positive values) region, and a central undecided region. Watermark bits are encoded only in the high-value tail regions, which are statistically more robust to perturbation. That is, because during the detection process, when DDIM inversion reconstructs the noise vector from a manipulated image, the introduced error is unlikely to be large enough to push a tail value from positive to negative or vice versa. Values near the center are so close to zero that even a small error flips their sign and corrupts the bit, which is why those positions are left unaltered, leaving the sampling noise less disrupted than prior methods.
Additionally, a two-stage hierarchical key encryption scheme is

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.

used to improve capacity: a static master key encrypts a randomly generated per-session key in stage one, and that session key encrypts the actual watermark bits in stage two.

6.4. Numerical Evaluation

Each of the three methods evaluates its performance using multiple evaluation metrics. As these methods are developed in the context of the technical standards available at the time of publication, the evaluation setups (e.g., the Stable Diffusion model, tested watermark attacks) and the chosen evaluation metrics vary between papers (e.g. only T2SM evaluates the impact of watermarking on image diversity). Consequently, a direct comparison of the individual evaluation results across all three methods is unreliable due to these methodological differences.
However, the authors of T2SM [3] provide a comprehensive evaluation of their method against several other watermarking approaches, including TR and GS, using the Stable Diffusion 3.5 Medium model, which employs a DiT architecture with a 16-channel latent space. This evaluation covers multiple metrics and enables a comparison of the methods under consistent conditions.
Table 2 provides both a high-level overview of the trade-offs between different watermarking properties and the quantitative results evaluated by the authors of T2SM for their method, GS and TR.
Accordingly, the values in Table 2 reflect only the findings reported by the authors of T2SM, as the research findings in the individual papers are not comparable. They highlight GS's and T2SM's superior imperceptibility and generative diversity compared to TR. This does not contradict Tree-Ring's foundational advantage in structural robustness against spatial and geometric attacks due to them prioritizing embedding in the frequency-domain [1] which is not represented in Table 2.

7. Discussion

The evaluation of watermark properties and functionalities (capacity, implementation and operation efficiency, imperceptibility and robustness) and their obstacles as well as current LDM watermarking methods presented in this paper reveals that robustness and implementation efficiency are the decisive parameters when it comes to real-world applicability.
The six types of stakeholder, whose interests ultimately define what a watermark should offer, expose why these two dimensions are uniquely prohibitive: Model owners are unlikely to adopt methods which might not be profitable because they degrade generation quality, are perceptible or impose significant computational overhead. Additionally, the model owner, the compliant user and the consumer presumably share an interest in a watermark having a high capacity because it offers concrete information so malicious and unauthorized actors can be held accountable and ownership can be proven. Yet those aspects become incidental, when it comes to a lack of robustness and I argue that a semi-robust watermark is suboptimal and not applicable. Detection unreliability through false-positive and -negative rates as well as removal attacks and forgery pose a major challenge for large-scale implementation of AI watermarks, not only in diffusion models, but in AI generated content in general.
Watermarks intended for copyright protection and legal application must meet an accountability threshold where false-positives and -negatives are vanishingly rare, because even occasional failures undermine trust and liability as well as enable false accusations.
Moreover, the same computational achievements that improve diffusion model quality simultaneously enhance adversaries capabilities, because the generation model is capable of both generation and impairing of the watermark. The vulnerability of the model architecture can negatively impact a watermark scheme's robustness performance and must be considered when evaluating robustness performance in future research. Furthermore, as the analysis of the initial noise altering watermarking methods suggests, the trade-off between parameters is moved less than resolved. As Table 2 summarizes, TR, GS and T2SM form a series in which each approach works on its predecessor's drawbacks at a cost elsewhere: TR's Fourier-domain pattern offers robustness to geometric and perturbation attacks but disrupts the Gaussian noise distribution, negatively impacting generation quality. GS and T2SM succeed to less disturb the initial noise distribution but give up the robustness improvement offered by working in the frequency domain like TR.
This dynamic shows that the progress on existing paradigms has not yet closed the deployment gap and suggests barriers which method-specific robustness improvements cannot overcome in isolation, implying a need to reasses current approaches while taking model robustness into account.

7.1. Limitations

This work aims to offer a short but broad and comprehensive overview by performing a high-level analysis of current technical obstacles and resolution approaches, and therefore intentionally is only marginally looking at evaluation metrics and is not performing an independent evaluation.
Additionally, I did not discuss legal regulation and standardization barriers which form a new category of obstacles. For watermarking to be held accountable, generative AI providers would need to adopt an international overarching scheme as well as clearly defined responsibility among the user types. Without coordination and clear ownership, detection is fragmented and liability unclear [5, 7, 19].

8. Conclusion

Diffusion models have significantly advanced the quality and accessibility of synthetic image generation and existing initial-noise watermarking methods hold considerable promise.
Nevertheless, this analysis indicates that the obstacles to adoption stem less from insufficient technical progress than from an unresolved conflict between the model architecture, which is used to generate, detect and potentially undermine the watermark, and the trade-offs between properties and stakeholder interests. As long as watermark robustness depends on the same diffusion model that can be used to remove it and detection remains semi-reliable, effective regulation of the consequences of synthetic image distribution remains unlikely.
This suggests, to enable large-scale deployment, approaches that do not rely solely on the generative model itself are worth investigating and future research and policy should treat watermark robustness as a structural requirement rather than one property among several.

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.

References

[1] Y. Wen, J. Kirchenbauer, J. Geiping, and T. Goldstein, "Tree-rings watermarks: Invisible fingerprints for diffusion images," in *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. [Online]. Available: https://openreview.net/forum?id=Z57JrmubNl
1, 2, 3, 4, 6

[2] Z. Yang, K. Zeng, K. Chen, H. Fang, W. Zhang, and N. Yu, "Gaussian shading: Provable performance-lossless image watermarking for diffusion models," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 12 162–12 171.
1, 2, 3, 4, 5

[3] J. Yang, H. Fang, W. Zhang, N. Yu, and K. Chen, "T2smark: Balancing robustness and diversity in noise-as-watermark for diffusion models," in *Advances in Neural Information Processing Systems 38*. Neural Information Processing Systems Foundation, Inc. (NeurIPS), 2025, pp. 131 526–131 552. [Online]. Available: https://doi.org/10.52202/085713-3958
1, 2, 3, 4, 5, 6

[4] J. Oppenlaender, J. Silvennoinen, V. Paananen, and A. Visuri, "Perceptions and realities of text-to-image generation," in *26th International Academic Mindtrek Conference*. ACM, 2023, pp. 279–288. [Online]. Available: https://doi.org/10.1145/3616961.3616978
1

[5] X. Zhao, S. Gunn, M. Christ, J. Fairoze, A. Fabrega, N. Carlini, S. Garg, S. Hong, M. Nasr, F. Tramer, S. Jha, L. Li, Y.-X. Wang, and D. Song, "Sok: Watermarking for ai-generated content," in *2025 IEEE Symposium on Security and Privacy (SP)*, 2025, pp. 2621–2639.
1, 2, 3, 4, 6

[6] H. Luo, L. Li, and J. Li, "Digital watermarking technology for ai-generated images: A survey," *Mathematics*, vol. 13, p. 651, 2025. [Online]. Available: https://doi.org/10.3390/math13040651
1, 2, 3, 4

[7] K. Ren, Z. Yang, L. Lu, J. Liu, Y.-M. Li, J. Wan, X. Zhao, X. Feng, and S. Shao, "Sok: On the role and future of aigc watermarking in the era of gen-ai," *ArXiv*, vol. abs/2411.11478, 2024. [Online]. Available: https://api.semanticscholar.org/CorpusID:274130796
1, 2, 3, 4, 6

[8] X. Zhao, K. Zhang, Z. Su, S. Vasan, I. Grishchenko, C. Kruegel, G. Vigna, Y.-X. Wang, and L. Li, "Invisible image watermarks are provably removable using generative ai," in *38th Conference on Neural Information Processing Systems (NeurIPS 2024)*, 2024. [Online]. Available: https://arxiv.org/abs/2306.01953
1, 2, 3

[9] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proceedings of the 34th International Conference on Neural Information Processing Systems*, ser. NIPS '20. Red Hook, NY, USA: Curran Associates Inc., 2020.
1, 2

[10] P. Fernandez, G. Couairon, H. Jégou, M. Douze, and T. Furon, "The stable signature: Rooting watermarks in latent diffusion models," in *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 22 409–22 420.
2

[11] H. Ci, P. Yang, Y. Song, and M. Z. Shou, "Ringid: Rethinking tree-ring watermarking fornbsp;enhanced multi-key identification," in *Computer Vision – ECCV 2024: 18th European Conference, Milan, Italy, September 29–October 4, 2024, Proceedings, Part XXVIII*. Berlin, Heidelberg: Springer-Verlag, 2024, p. 338–354. [Online]. Available: https://doi.org/10.1007/978-3-031-73390-1_20
2

[12] Y. Chen, J. Vice, N. Akhtar, N. Haldar, and A. Mian, "Image watermarking of generative diffusion models." arXiv, WorkingPaper, 2025.
2, 3, 4

[13] O. Ronneberger, P.Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, ser. LNCS, vol. 9351. Springer, 2015, pp. 234–241, (available on arXiv:1505.04597 [Add to Citavi project by ArXiv ID] [cs.CV]). [Online]. Available: http://lmb.informatik.uni-freiburg.de/Publications/2015/RFB15a
2

[14] W. S. Peebles and S. Xie, "Scalable diffusion models with transformers," *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 4172–4182, 2022. [Online]. Available: https://api.semanticscholar.org/CorpusID:254854389
2

[15] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *International Conference on Learning Representations*, 2021. [Online]. Available: https://openreview.net/forum?id=St1giarCHLP
2, 4

[16] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. [Online]. Available: https://github.com/CompVis/latent-diffusionhttps://arxiv.org/abs/2112.10752
2

[17] K. Arabi, B. Feuer, R. T. Witter, C. Hegde, and N. Cohen, "Hidden in the noise: Two-stage robust watermarking for images," in *2025 International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: https://arxiv.org/abs/2412.04653
2, 3, 4, 5

[18] Y. Song, J. N. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," *ArXiv*, vol. abs/2011.13456, 2020. [Online]. Available: https://api.semanticscholar.org/CorpusID:227209335
3

[19] B. Rijsbosch, G. van Dijck, and K. Kollnig, "Adoption of watermarking for generative ai systems in practice and implications under the new eu ai act," 2025. [Online]. Available: https://arxiv.org/abs/2503.18156

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.

AI Acknowledgment

I acknowledge the use of *GPT-4o* by *OpenAI* accessed through the platform *SciSpace* to improve organization in the form of targeted search and collecting of suitable research papers as well as sorting them by topic. Furthermore it was used to refine language and better text flow as well as correcting passive tense. This happened always

Claude Sonnet 5.5 by *Anthropic* was used to support the structural coding of the tables and to replace my manually defined citations with the corresponding BibTeX citation keys in LaTeX.

All AI tasks were continuously monitored and revised by the author.

© 2026 The Author(s)
Academic Conference Simulation Proceedings © 2026 The Filmuni Association.